Replication of Study 2 by Jamie Amemiya, Gail D. Heyman, and Caren Walker (2024, Cognitive Science)

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Introduction

Study 2 of Amemiya et al. (2024) finds that adults who wanted to persuade others to support or reject a public policy selectively presented comparison cases that encourage counterfactual inferences consistent with their position. Specifically, participants in the *Motivated to Support* condition were more likely to show the supporting comparison cases than participants in the *Motivated to Reject* condition, and that participants in the *Motivated to Reject* condition

were more likely to show the rejecting comparison cases than participants in the *Motivated to Support* condition.

I chose this study because I'm interested in the development of our ability to understand that the way evidence was selected affects how much weight we should give it. Past work has shown that young children are sensitive to how samples are generated (Kushnir et al., 2010) and can strategically select true evidence to deceive others (Rhodes et al., 2015). These studies, along with the target study, primarily approach the problem from the perspective of the person producing or selecting evidence, but I aim to also investigate the less-studied child-as-consumer side and to examine the relation between these two sides of the problem: 1) understanding how to select evidence to persuade someone, and 2) recognizing that the process by which true evidence is selected affects the value of that evidence (how informative it is). This project would provide an important foundation for my future work by establishing whether the evidence-selection behavior observed in adults can be replicated before extending the paradigm to investigate how children might produce and evaluate selectively presented evidence.

Participants will read a vignette about a novel virus spreading in five regions: Hawaii (US), Hokkaido (Japan), Te Waipounamu (New Zealand), Oregon (US), and California (US). In this vignette, leaders were unsure how to respond, so they designed a set of policies ("Policy A"). Only Hawaii implemented Policy A, on Day 5. The other four regions waited to see whether it worked. Participants will be randomly assigned to imagine being a politician who either strongly believes Policy A would be effective (*Motivated to Support* condition), or strongly believes Policy A would be ineffective (*Motivated to Reject* condition). Participants will then view a full five-location graph and describe the data in their own words. The graph is a figure that plots new virus cases over Days 0-10 for all five regions, with a vertical line at Day 5 marking the intervention. Hawaii's cases are shown to drop sharply right after Day 5, so there was strong temporal covariation between policy and decline in every condition. Oregon and California kept climbing after Day 5, implying the supporting counterfactual inference that Hawaii's cases would have continued to rise without the policy. Hokkaido and Te Waipounamu declined on their own after Day 5, implying the rejecting counterfactual inference that Hawaii's cases would have fallen even without the policy.

For the primary outcome, participants will choose one of the four following graphs to present on the nightly news in support of their assigned position: (a) Target only: Hawaii's trajectory alone (b) Target + Supporting Comparisons: Hawaii plus Oregon and California (c) Target + Rejecting Comparisons: Hawaii plus Hokkaido and Te Waipounamu (d) Target + All Comparisons: all five regions Participants will provide an open-ended explanation for their choice. I will test whether participants selectively present comparisons consistent with their assigned position. I will also examine whether the emphasized dimension of the data interacts with experimental condition, following the authors' approach.

The authors implemented the survey using Qualtrics, but I will be using jsPsych. I will use the authors' publicly available materials as a starting point, adapting them as needed. Participants will complete attention, comprehension, and manipulation checks throughout the task.

Anyone who fails a comprehension check will be excluded, and I will preregister the exclusion criteria before data collection. One potential challenge is participant attrition or exclusion due to failed comprehension checks, particularly because the task includes training modules, corrective feedback, and open-ended responses. To account for this, I will recruit approximately 20% more participants than the target analyzable sample to account for exclusions, using the exclusion rate reported in the original study (approximately 14%) as an estimate. Because the original study provides a useful benchmark for anticipating exclusions, oversampling should help with securing an adequate final sample. Furthermore, if very few participants select certain graphs in some experimental conditions, logistic regression may not provide reliable estimates. If this is the case, I will use a Chi-square test to see whether choosing a particular option (e.g., Target + Rejecting Comparisons) vs. not (any other option) varies by condition. In addition, I will test whether the emphasized dimension (islands vs. U.S. states) predicts participants' causal judgments and whether it interacts with experimental condition, as in the original study.

Beyond replicating the original study, I am interested in using the extra time available to begin investigating how people evaluate evidence selected by others. After participants choose a graph to present, they will be presented with a graph chosen by another politician who supports or opposes Policy A and be asked how convincing they find the politician's argument. This would examine whether people recognize that evidence can be factually accurate yet selectively presented to support a particular conclusion, and whether their evaluations are sensitive to the selection process rather than just the evidence itself.

Through this project, I hope to gain both technical skills in implementing behavioral experiments and a deeper understanding of how to make thoughtful and adaptive analytical decisions. I have primarily used Qualtrics to build studies and have little hands-on experience with jsPsych, so I am excited to learn how to use it to build an interactive behavioral task and implement experimental counterbalancing. I also hope to become more comfortable navigating the analytical decisions that arise when working with real data, such as determining which analyses are most appropriate for different research questions and the data distribution you get. I want to improve my understanding of when the assumptions of a planned analysis may be violated and when additional and/or alternative statistical tests are warranted. For example, I am interested in learning how to anticipate and address potential analytical issues. Broadly, I hope this project will help equip me with the technical independence to implement behavioral studies and the statistical judgment to make rigorous, well-justified decisions throughout the research process.

See the project repository here and the original paper here.

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.